

# Cosmic Microwave Background temperature measurement

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## Abstract

We reproduced the landmark 1964 experiment of Penzias and Wilson that led to the discovery of the Cosmic Microwave Background (CMB), employing modern instrumentation and electronics. Using a horn antenna pointed at the sky, we measured the received power and interpreted it in terms of an equivalent temperature. Since the detected signal contains contributions from both the CMB and the atmosphere, we varied the zenith angle to characterize the atmospheric contribution. By extrapolating to zero air mass, we obtained an estimate of the CMB temperature  $-11.5 \pm 22.0$  K at  $1\sigma$  confidence level.

## 1 Introduction

The goal of our experiment is to measure the temperature of the Cosmic Microwave Background through ground-based measurements. The CMB radiation is the first light emitted in the Universe at the time of recombination, around 380,000 years after the Big Bang. It exhibits a blackbody spectrum at the temperature of 2.73 K [?], with spectral intensity rising with frequency in the Rayleigh-Jeans regime. Detecting this signal from the ground is particularly challenging, since the atmospheric emission can largely exceed and obscure the weak CMB signal. An operating frequency of 20 GHz was thus selected as a fair compromise between enhancing the CMB signal and limiting the effects of atmospheric opacity, which become more significant at higher frequencies.

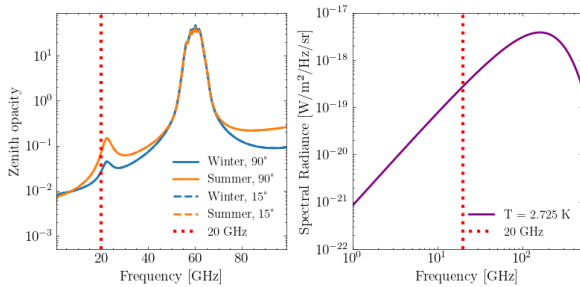


Figure 1: CMB blackbody spectrum and atmospheric opacity as a function of frequency.

In order to discriminate the effect of the atmosphere from the CMB radiation, it is important to account for the airmass, i.e. the effective path length through the atmosphere, which scales with the zenith angle  $\theta$  as  $\frac{1}{\cos \theta}$ .

The dependence of the measured temperature on  $\theta$  can be derived from the radiative transfer equation. Considering a uniform atmosphere characterized by a temperature  $T_{\text{atm}}$  and assuming that the optical depth at the zenith is given by  $\tau \ll 1$ , the temperature measured by the antenna, as a function of  $\theta$ , reads

$$T(\theta) = T_{\text{CMB}} \left( 1 - \frac{\tau}{\cos \theta} \right) + T_{\text{atm}} \frac{\tau}{\cos \theta}, \quad (1)$$

where we indicate the CMB temperature by  $T_{\text{CMB}}$ .

The link between the measured power and the temperature estimate that we want to extract is given by Planck's law. Indeed, we assume that every emitting object that we observe through the antenna is a uniform black body, characterized by an energy density  $\rho$  that is

$$\rho = \frac{dE}{dV d\nu} = \frac{8\pi\nu^2}{c^3} \frac{h\nu}{e^{h\nu/kT} - 1}. \quad (2)$$

To estimate the received power per unit frequency, we assume detailed balance: in thermal equilibrium, the power radiated by the antenna equals the power it receives, i.e.  $P_{\text{rec}} = P_{\text{rad}}$ . Under this condition, the antenna temperature can be directly associated with the temperature of the observed radiation field.

Thus, we find a linear relation between the power received by the antenna and the temperature of the emitting source

$$\frac{dP_{\text{rec}}}{d\nu} = \frac{h\nu}{e^{h\nu/kT} - 1} \approx \frac{h\nu}{1 + \frac{h\nu}{kT} - 1} = kT, \quad (3)$$

while the total power collected by the antenna can be obtained by integration over the selected frequency range  $\Delta\nu$

$$P_{\text{rec}} = kT\Delta\nu. \quad (4)$$

<sup>1</sup>  $\frac{h\nu}{kT} \approx [0.003 - 0.1]$  at 10 – 300 K and 20 GHz.

## 2 Methodology

### 2.1 Experimental setup

We assemble the following setup:

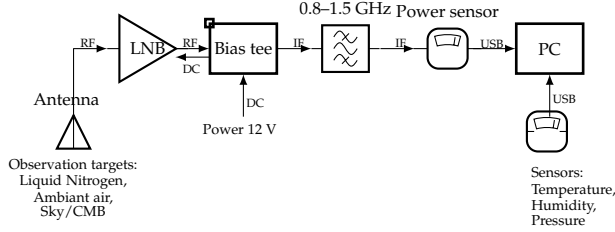


Figure 2: Main setup for our measurements.

A conical horn antenna is used to collect radiation from the sky and from laboratory reference sources. The antenna is attached to a graduated turntable to vary the direction of pointing. The antenna output is connected to the Low-Noise Block downconverter (LNB) VK2L from Labgear, which is sensitive to frequencies in [19.7, 20.4] GHz. It has a local oscillator of 21.2 GHz, which produces an output signal in [0.8, 1.5] GHz via the heterodyne technique, and we add filters to ensure that the signal we study is only included in this range. This signal was finally recorded thanks to the power sensor PWR-4RMS from Mini Circuits.

As only the power received from the sky can be measured, the relation given in Eq. 1 must be determined through calibration. During this process, the power from two reference sources with known temperatures is measured by the sensors, allowing the establishment of a linear calibration curve that converts the measured sky power into an equivalent temperature.

### 2.2 Calibration

During the measurements, the horn antenna is successively pointed towards two reference loads of known temperature. This procedure establishes an experimental calibration curve that links the measured power to the corresponding radiation temperature.

In our experiment, conversion from antenna temperature to measured power is parameterized as

$$P = G \times (T + T_{\text{sys}}), \quad (5)$$

where  $G$  is the system gain and  $T_{\text{sys}}$  is the system temperature, representing noise added by the receiver electronics.

This experiment depends on ambient conditions, such as temperature, humidity, and pressure, which can fluctuate on short timescales, potentially affecting the measurements. To mitigate these variations,

each sky observation is accompanied by a simultaneous calibration (on-site calibration) with two reference loads: a foam block at ambient temperature and a foam block cooled with liquid nitrogen. This ensures that, for each pointing angle, updated temperature–power linear coefficients are available. Moreover, this allows us to determine the system gain  $G$

$$G = \frac{P_w - P_c}{T_w - T_c}, \quad T_{\text{sys}} = \frac{T_w P_c - T_c P_w}{P_w - P_c}, \quad (6)$$

where  $T_c$  and  $T_w$  are the temperatures of the cold and warm reference loads, respectively.

The uncertainties in  $\sigma_G$  and system temperature  $\sigma_{T_{\text{sys}}}$  were propagated from the measurement uncertainties in power ( $\delta P_i$ ) and load temperature ( $\delta T_i$ ) as follows:

$$\sigma_G^2 = \frac{1}{\Delta T^2} (\delta P_w^2 + \delta P_c^2) + \frac{G^2}{\Delta T^2} (\delta T_w^2 + \delta T_c^2), \quad (7)$$

$$\sigma_{T_{\text{sys}}}^2 = \frac{(P_c^2 \delta P_w^2 + P_w^2 \delta P_c^2)}{G^4 \Delta T^2} + \frac{(\delta T_w^2 + \delta T_c^2)}{G^2 \Delta T^2},$$

where  $\Delta T = T_w - T_c$  is the temperature difference between the warm and cold loads.

To further characterize the power and temperature relation, additional measurements are performed. A foam block equipped with temperature sensors is placed on the surface of liquid nitrogen and its gradual warming is monitored. During this process, both the temperature of the load and the corresponding power measured by the antenna are recorded (off-site calibration).

### 2.3 Sky Measurements

The antenna measures the power from the sky at different zenith angles. The angles  $\theta$  are chosen such that the corresponding airmass is sampled in increments of 0.25. Using the calibrated values of  $G$  and  $T_{\text{sys}}$  for each zenith angle, the measured power  $P_{\text{mes}}$  is converted into the corresponding sky temperature  $T_{\text{mes}}$  via Eq. 5, with a total uncertainty given by

$$\sigma_{T_{\text{mes}}}^2 = \frac{(\delta P_{\text{mes}})^2}{G^2} + \frac{P_{\text{mes}}^2}{G^4} \sigma_G^2 + \sigma_{T_{\text{sys}}}^2. \quad (8)$$

Two complete scans were performed: a forward scan from  $0^\circ$  to  $70^\circ$  and a backward scan from  $70^\circ$  back to  $0^\circ$  to account for possible system drifts. At each angle, 20–30 measurements of the sky, cold load, and warm load signals were acquired over 12–18 s using an Arduino system, with a sampling interval of 0.6 s. The statistical and systematic uncertainties associated with these measurements are detailed in the following section.

The resulting sky temperatures are expected to vary linearly with airmass as in Eq. 1. Assuming a

typical zenith optical depth of  $\tau = 0.1$  at 20 GHz under clear skies, the measured sky temperatures are fitted as a function of airmass. Extrapolating this fit to zero airmass yields our estimate of the CMB temperature,  $T_{\text{CMB}}$ , while the atmospheric temperature,  $T_{\text{atm}}$ , is also extracted.

### 3 Results

#### 3.1 Error Analysis

**Statistical uncertainties:** The statistical uncertainty associated with each measurement of different angles is estimated from the standard deviation  $\sigma$  of the recorded values:

$$\sigma = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (X_i - \bar{X})^2}, \quad (9)$$

where  $X_i$  denotes the  $i$ -th measurement,  $\bar{X}$  is the mean, and  $N$  is the total number of samples.

**Systematic uncertainties:** Several sources contribute to systematic uncertainties in our measurements:

- *Angle:* The rotation system precision ( $\approx 1^\circ$ ) combined with environmental disturbances (wind, mechanical adjustments) leads to a conservative angular uncertainty of  $2^\circ$ , propagating to airmass as  $\sigma_x = \frac{\cos \theta}{\sin^2 \theta} \sigma_\theta$ .
- *Power:* A systematic uncertainty of 0.1 dBm was assigned to the power measurements, accounting for temporal gain drift observed when measuring the cold source where the temperature of the source is most stable (see Fig. 3).
- *Temperature:* the recording resolution of 0.1 K of our sensors defines the systematic uncertainty for temperature measurements.

| Measurement       | $\sigma_{\text{stat}}^{\text{mean}}$ | $\sigma_{\text{sys}}$ |
|-------------------|--------------------------------------|-----------------------|
| Power             | 0.01 dBm                             | 0.1 dBm               |
| $T_{\text{cold}}$ | 0.01 K                               | 0.1 K                 |
| $T_{\text{warm}}$ | 0.04 K                               | 0.1 K                 |
| $\theta$          | -                                    | $2^\circ$             |

Table 1: Summary of systematic and mean statistical uncertainties for angle, power and temperature measurements.

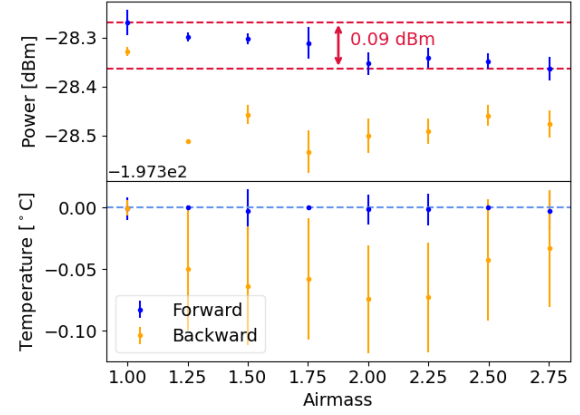


Figure 3: Systematic error estimation from the stable liquid nitrogen reference.

#### 3.2 Calibration

Following the method introduced in Sec. 2.2, the system gain  $G$  and system temperature  $T_{\text{sys}}$  were calibrated for each zenith angle. The results are presented in Fig. 4. The data reveal a clear anti-correlation between  $G$  and  $T_{\text{sys}}$ , suggesting an underlying drift in receiver performance. Ideally, a stable system would show no systematic relation between gain and noise. Furthermore, the forward scan exhibits greater stability in both parameters compared to the backward scan, indicating that system drift accumulated over the duration of the measurement sequence.

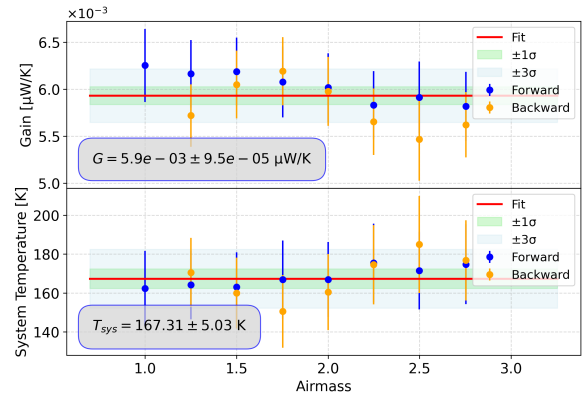


Figure 4: System gain and system temperature as a function of airmass.

#### 3.3 Extraction of the CMB Temperature

The result of the linear fit is shown in Fig. 5. All three fits yield non-physical values for  $T_{\text{CMB}}$ , with an uncertainty  $1\sigma$  an order of magnitude larger than the expected signal. The forward and backward scans show

a clear systematic drift:  $T_{\text{CMB}}$  decreases while  $T_{\text{atm}}$  increases from the forward to the backward scan. The measurement places a  $3\sigma$  upper limit of  $T_{\text{CMB}} < 54.4$  K.

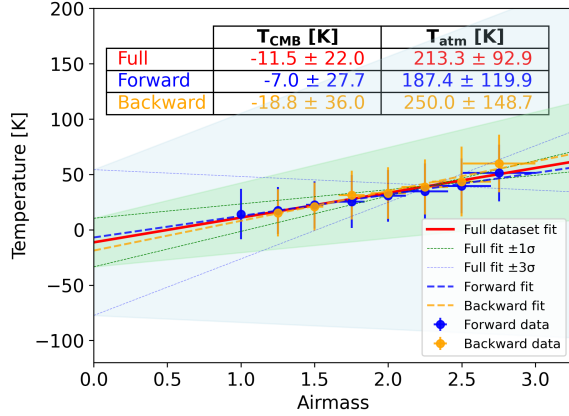


Figure 5: Estimation of the CMB Temperature

## 4 Discussion

### 4.1 Systematic errors

The derived  $3\sigma$  upper limit of  $T_{\text{CMB}}$  is consistent with the established value of 2.73 K but lacks astrophysical significance. The measurement is fundamentally limited by systematic errors (see Table 1) and gives an unphysical best-fit  $T_{\text{CMB}}$  and  $T_{\text{atm}}$ . The parameters  $\tau$  and  $T_{\text{atm}}$  are degenerate in our fit, which only constrains their product. Choosing an optical depth of  $\tau = 0.074$  yields a realistic  $T_{\text{atm}} \approx 20^\circ\text{C}$  for the full fit, demonstrating this degeneracy. However, this adjustment cannot resolve the negative value  $T_{\text{CMB}}$ , which combined with the significant drift observed, points to dominant instrumental instabilities rather than atmospheric modeling as the primary source of error.

Systematic effects in the power measurement system dominate the error budget, particularly the temporal gain drift that could depend on LNB temperature, humidity, or pressure variations. Although calibration was performed simultaneously with measurements, the finite antenna rotation time (several seconds) may have introduced additional variability. The potential ground pickup from the antenna side lobes represents another source of contamination. These combined effects currently prevent the precise determination of the CMB temperature with our experimental setup.

### 4.2 Calibration

We further investigate the relationship between power and temperature during the off-site calibration.

Figure 6 shows a non-linear correlation between the received power and temperature during liquid nitrogen evaporation. This deviation from the expected linear response is attributed to thermal and electromagnetic non-idealities. Thermally, significant time lags in sensor response and temperature gradients across the foam occur due to finite thermal conductivity during rapid warming. Electromagnetically, the foam absorber used in our experiment may not behave as a perfect blackbody at 20 GHz. Its emissivity is likely non-uniform and temperature-dependent, causing deviations from ideal blackbody behavior through internal reflections and frequency-dependent absorption.

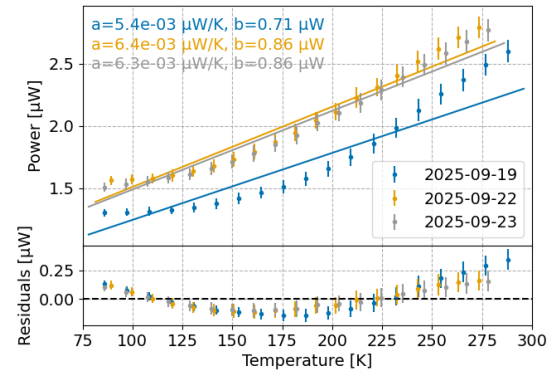


Figure 6: The parameters  $a$  and  $b$  represent the slope and intercept, respectively, of the linear function  $y = ax + b$  fitted to the power versus temperature data.

## 5 Conclusions and Outlook

Our experiment successfully demonstrated the principles of radiometric detection and established a  $3\sigma$  upper limit of  $T_{\text{CMB}} < 54.4$  K. However, we were unable to make a precise measurement of the CMB temperature due to large systematic errors.

The main challenge was instrumental instability. We observed a significant drift in the system's gain and noise temperature between measurements, and our calibration revealed that the relationship between power and temperature was not perfectly linear. This shows our reference loads were not ideal blackbodies. Together, these effects created uncertainties much larger than the CMB signal we were trying to detect, leading to a non-physical best-fit value.

Despite these challenges, it is clear that detecting a weak cosmic signal requires careful control of the instrument itself. For future work, using a Dicke switch or calibration loads made of near-ideal 20 GHz blackbody materials would improve measurement accuracy.

## References

- [1] D. J. Fixsen, E. S. Cheng, J. M. Gales, J. C. Mather, R. A. Shafer, and E. L. Wright, “The Cosmic Microwave Background Spectrum from the Full COBE FIRAS Data Set,” *The Astrophysical Journal*, vol. 473, no. 2, pp. 576–587, Dec. 1996. <https://arxiv.org/abs/astro-ph/9605054>
- [2] International Telecommunication Union, *Recommendation ITU-R P.676-13: Attenuation by atmospheric gases*, 2022. <https://www.itu.int/rec/R-REC-P.676-13-202202-I/en>